Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 11: Transportation in Animals and Plants

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	11 — Transportation in Animals and Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps target the 'blood is blue in veins' myth, the artery-vs-vein direction and pressure flips, the 'veins have valves because of high pressure' reversal, the atria-vs-ventricles wall-thickness mix-up, the pulse-is-a-valve-snap error, the 'urea is a useful food' and 'urea made in the kidney/bladder' confusions, the xylem-vs-phloem cargo and direction swap, the 'transpiration stops in heat' and 'wilting means poisoned soil' misreads, and the diffusion-is-enough-for-big-bodies slip. Do not write on this paper — mark answers on your answer sheet.

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1. A doctor reports that a patient's blood, after a serious cut, takes far too long to stop bleeding, yet the patient's oxygen levels and ability to fight germs are normal. Which blood component is most likely too low?

(A) red blood cells
(B) white blood cells
(C) plasma
(D) platelets

2. Anaemia is a condition in which a person tires easily and gets breathless because tissues are starved of oxygen. The blood feature most directly behind anaemia is:

(A) too few platelets, so the blood cannot clot
(B) too few white cells, so infections spread
(C) too little haemoglobin in the red cells to carry enough oxygen
(D) too much plasma, making the blood watery

3. Plasma makes up a large part of blood. Besides carrying the blood cells, plasma's role includes:

(A) transporting dissolved substances such as digested food, wastes like urea, and other materials

(B) producing haemoglobin to colour the blood red

(C) engulfing and destroying invading bacteria

(D) sealing wounds by forming a solid clot

4. A microscope slide of blood shows three kinds of cells. The most numerous are small, round and have no nucleus; these are red cells. The fact that mature red cells lack a nucleus is useful because it:

(A) lets the cell fight germs more aggressively

(B) leaves more room inside the cell for haemoglobin, so each cell can carry more oxygen

(C) allows the cell to form a clot when needed

(D) makes the cell able to make its own food

5. During a bacterial infection a person's white-cell count rises sharply. The best explanation for this rise is that white cells:

(A) carry the extra oxygen the sick body suddenly needs

(B) are the body's defenders, so the body makes more of them to fight the invading germs

(C) are needed in large numbers to clot the blood at the infection site

(D) turn red and become red cells once the germs are killed

6. Match each task to its blood component: (i) carry oxygen, (ii) defend against disease, (iii) help blood clot. The correct match is:

(A) (i) red cells, (ii) white cells, (iii) platelets

(B) (i) white cells, (ii) platelets, (iii) red cells

(C) (i) platelets, (ii) red cells, (iii) white cells(D) (i) plasma, (ii) red cells, (iii) white cells

7. A sample of blood is spun in a tube so the parts separate. The clear, pale-yellow liquid that rises to the top, with the cells packed below, is best identified as:

(A) plasma, the liquid part of the blood (B) haemoglobin, the red pigment

(C) urea, the waste filtered by kidneys (D) lymph drained from the tissues

8. A patient has lost a lot of blood and is given a transfusion to restore oxygen delivery to the tissues. The component of the donated blood that most directly restores oxygen delivery is the:

(A) platelets that help clotting (B) white blood cells that fight germs

(C) plasma proteins that fight infection (D) red blood cells with their haemoglobin

9. Why is iron in the diet important for healthy blood?

(A) iron forms the platelets that clot the blood

(B) iron is the liquid plasma in which cells float

(C) iron makes the white cells that fight germs

(D) iron is needed to build haemoglobin, which carries oxygen

10. Two blood samples are compared. Sample X is bright red; sample Y is duller, darker red. The most reasonable explanation within Class 7 science is that:

(A) X carries more oxygen bound to its haemoglobin, while Y carries less

(B) X has more platelets and Y has fewer(C) X is real blood and Y has turned into urine

(D) Y is blue blood that has not yet met air

11. An artery and a vein are cut in an accident. The artery spurts blood in strong bursts while the vein oozes in a steadier flow. The best reason for the difference is that the artery:

(A) carries blood back toward the heart slowly, so it spurts

(B) has valves that snap, causing the spurts

(C) is thinner than the vein, so blood escapes faster

(D) carries blood pumped directly from the heart at high pressure, so it spurts with each heartbeat

12. A vessel is examined and found to have valves along its length and relatively thin walls, carrying blood at low pressure. This vessel is a:

(A) artery (B) capillary

(C) vein (D) ureter

13. Why do veins, but not arteries, need valves?

(A) blood in veins moves at very high pressure and must be slowed by valves

(B) blood in veins moves at low pressure and could slip backward, so valves keep it moving toward the heart

(C) valves help arteries pump blood, so veins copy them

(D) valves in veins make the heartbeat sound louder

14. Capillaries are described as the 'exchange vessels' of the body. The structural feature that makes them suited to exchange is that their walls are:

(A) thick and muscular, to push substances out

(B) only one cell thick, so gases and substances pass through easily

(C) lined with many valves, to trap substances(D) made of cartilage, to filter the blood

15. Trace the general path of blood through the vessel types as it leaves the heart, supplies a muscle, and returns. The correct order is:

(A) vein → capillaries → artery (B) artery → vein → capillaries

(C) capillaries → artery → vein (D) artery → capillaries → vein

16. A student argues, 'All arteries carry oxygen-rich blood and all veins carry oxygen-poor blood.' The most accurate correction is:

(A) completely true with no exceptions at all

(B) completely false; arteries always carry oxygen-poor blood

(C) usually true, but the artery to the lungs carries oxygen-poor blood and the vein from the lungs carries oxygen-rich blood

(D) true only for children, not adults

17. Why are an artery's walls thick, elastic and muscular compared with a vein's?

(A) they must hold valves that veins lack

(B) they carry blood very slowly, so need extra padding

(C) they must stretch and withstand the high pressure of blood pumped straight from the heart

(D) they are filtering the blood as it passes

18. In which vessel type is blood pressure HIGHEST and in which is flow SLOWEST?

(A) highest in veins; slowest in arteries (B) highest in arteries; slowest in capillaries

(C) highest in capillaries; slowest in veins(D) highest in arteries; slowest in arteries

19. Through the skin of the forearm, some vessels look bluish. A classmate concludes the blood inside is blue. The scientific truth is:

(A) the blood really is blue until it meets air at a wound

(B) only oxygen-poor blood is genuinely blue

(C) the blood is dark red; it only looks bluish because of how light passes through the skin

(D) the vessels are filled with blue plasma

20. Which single feature is found in arteries but NOT in typical veins?

(A) valves that prevent blood flowing backward(B) a wall only one cell thick

(C) the job of carrying blood toward the heart

(D) thick, muscular, elastic walls built to take high pressure

21. The human heart has four chambers. Why is having separate chambers for oxygen-rich and oxygen-poor blood an advantage?

(A) it makes the heartbeat sound louder

(B) it lets the heart beat without using any muscle

(C) it allows the blood to change colour faster

(D) it keeps the two kinds of blood from mixing, so the body gets fully oxygenated blood

22. The two upper chambers of the heart are the atria and the two lower are the ventricles. Compared with the atria, the ventricles have thicker muscular walls mainly because they:

(A) must pump blood out of the heart with more force

(B) receive blood gently arriving from the veins(C) store urine before it leaves the body

(D) contain the valves that make the pulse

23. The 'lub-dub' sounds a doctor hears with a stethoscope are produced by:

(A) blood squeezing through the thin capillaries

(B) the heart's valves closing as blood is pumped

(C) the lungs drawing in air with each breath(D) the pulse beating at the wrist

24. Each heartbeat sends a surge of blood into the arteries, and this can be felt as a throb at the wrist. This throb is the:

(A) pulse, caused by the artery expanding with each beat
(B) clotting of the blood at the skin (C) valves of the veins snapping shut
(D) lungs pushing air down the arm

25. A child's heart beats 90 times per minute. In a quarter of a minute (15 seconds), how many beats — and hence pulses — would you expect?

(A) about 90 beats (B) about 45 beats
(C) about 22 or 23 beats (D) about 6 beats

26. A nurse counts 24 pulse beats in 20 seconds. The pulse rate per minute is:

(A) 24 beats per minute (B) 48 beats per minute
(C) 72 beats per minute (D) 120 beats per minute

27. After sprinting, Ravi's heart beats much faster than at rest. The best reason is that his working muscles now need:

(A) less oxygen, so the heart speeds up to slow the blood
(B) to stop clotting, which only fast blood prevents
(C) to cool down by pumping blood more slowly
(D) more oxygen and food delivered quickly, and more waste carried away

28. A resting adult is found to have a pulse of 70 beats per minute, while a second resting adult shows 130. Compared with a typical resting rate of about 72, the second adult's rate is:

(A) exactly normal for rest (B) much higher than the usual resting rate
(C) lower than the usual resting rate (D) impossible in a living person

29. Which statement about the heart is correct?

(A) it is a muscular organ that contracts rhythmically to pump blood through vessels
(B) it is made mostly of bone that snaps shut to push blood
(C) it filters wastes out of the blood like a kidney
(D) it makes the body's food and sends it through the phloem

30. The kidneys remove a harmful nitrogen-containing waste from the blood and pass it out in urine. This waste is:

(A) glucose (B) oxygen
(C) urea (D) haemoglobin

31. Inside each kidney are about a million microscopic filtering units. These units are called:

(A) nephrons
(B) stomata
(C) platelets
(D) ventricles

32. Put the urinary structures in the order urine actually travels through them after being formed:

(A) kidney → urethra → bladder → ureter(B) bladder → ureter → kidney → urethra
(C) ureter → kidney → urethra → bladder
(D) kidney → ureter → urinary bladder → urethra

33. On a hot day a person sweats a great deal. Besides cooling the body, sweating also serves excretion because sweat:

(A) takes in oxygen through the skin
(B) removes some water and dissolved salts from the body
(C) carries food to the body cells
(D) thickens the blood so it clots faster

34. If a person's kidneys fail completely, harmful wastes build up in the blood. A machine can take over the cleaning. This treatment is called:

(A) dialysis
(B) transpiration
(C) transfusion
(D) circulation

35. Why must the body continually remove urea rather than store it?

(A) urea is a valuable food the body wants to keep
(B) urea is needed to colour the blood red(C) urea helps the blood to clot at wounds
(D) urea is toxic, so letting it build up would poison the body

36. Trace the journey of urea correctly from where it forms to where it leaves the body:

(A) made in the kidneys → carried by xylem → leaves through stomata
(B) made in body cells → carried by the blood → filtered out by the kidneys → leaves in urine
(C) made in the bladder → carried by arteries → leaves through the lungs
(D) made in urine → carried by ureters → stored in the heart

37. Both sweating and urine formation are examples of excretion because both:

(A) bring oxygen into the body
(B) deliver digested food to the cells
(C) pump blood around the body
(D) remove waste materials and excess water from the body

38. Plants do not have kidneys, yet they get rid of some wastes. One common way a plant rids itself of waste is by:

(A) storing waste in leaves that later fall off
(B) passing urea down a ureter into a bladder(C) filtering blood through plant nephrons
(D) sweating salts out through root hairs

39. Gums and resins seen oozing from tree trunks are best understood as:

(A) food the plant is sending to its roots (B) water being pulled up to the leaves

(C) waste products a plant stores or releases(D) oxygen the plant breathes out at night

40. Some gaseous wastes and excess water vapour leave a plant through tiny pores on the leaves. These pores are the:

(A) root hairs (B) stomata

(C) nephrons (D) platelets

41. Compared with animals, plant excretion is generally:

(A) faster, because plants have many kidneys

(B) done entirely through a heart and vessels

(C) impossible, because plants make no waste at all

(D) slower and simpler, since plants make less harmful waste and can store some of it

42. Which statement about how plants handle waste is correct?

(A) plants release some waste through stomata and store some in parts like leaves, bark, gums and resins

(B) plants filter all their waste through a pair of kidneys

(C) plants send all their waste up the xylem and out the roots

(D) plants have no waste of any kind to remove

43. Water and dissolved minerals absorbed from the soil are carried UP to the leaves of a plant through the:

(A) phloem (B) stomata

(C) xylem (D) root hairs

44. Root hairs are especially good at absorbing water and minerals because they:

(A) are green and make food for the root(B) pump water upward using tiny muscles

(C) greatly increase the surface area of the root in contact with soil water

(D) carry food down from the leaves

45. As water evaporates from the leaves, it creates a 'pull' that draws more water up the stem. This driving force is the:

(A) heartbeat of the plant (B) transpiration pull

(C) clotting pressure (D) phloem push from the roots

46. On a hot, dry, windy afternoon, transpiration from a plant's leaves will generally:

(A) stop entirely, since heat seals the leaves

(B) increase, so water is pulled up the stem faster

(C) reverse, pushing water down into the soil

(D) have no effect on water movement at all

47. A potted plant stands in moist soil yet wilts on a very hot afternoon. The best explanation is that:

(A) the soil water has suddenly become poisonous

(B) the leaves are losing water by transpiration faster than the roots can supply it

(C) the xylem has turned into phloem (D) the plant has stopped making food forever

48. A plant with most of its root hairs destroyed would MOST directly struggle to:

(A) make food in its leaves (B) lose water through its stomata

(C) absorb water and minerals from the soil(D) transport food down its phloem

49. If the xylem in a stem were completely blocked, the part of the plant to suffer FIRST would be the:

(A) leaves, because food can no longer reach them

(B) roots, because they would lose their absorbed minerals

(C) leaves, because water can no longer be carried up to them

(D) roots, because food made above could not reach them

50. Food made in the leaves by photosynthesis is carried to all other parts of the plant — including downward to the roots — by the:

(A) xylem (B) stomata

(C) root hairs (D) phloem

51. Unlike water in the xylem, food in the phloem can travel:

(A) only straight up to the topmost leaves(B) only straight down to the roots

(C) in more than one direction — up or down to wherever food is needed

(D) nowhere; food does not actually move in plants

52. A ring of bark — including the phloem — is stripped right around a tree trunk. The part of the tree most likely to starve is the:

(A) region below the ring, because food from the leaves can no longer travel down to it

(B) region above the ring, because water can no longer travel up to it

(C) leaves, because they can no longer make food at all

(D) roots, because the xylem has been cut

53. Sugar made in a leaf must reach a growing root tip AND a developing fruit at the same time. This is possible because the phloem:

(A) only ever carries food upward to fruits(B) pushes all food into the xylem first

(C) moves food only when the plant is in sunlight

(D) can transport food in more than one direction simultaneously

54. Which pairing of plant tissue and its cargo and direction is correct?

(A) xylem carries food downward; phloem carries water upward

(B) both xylem and phloem carry only water upward

(C) xylem carries water and minerals upward; phloem carries food in multiple directions

(D) xylem carries oxygen; phloem carries urea

55. A single-celled organism like an Amoeba has no heart, blood vessels or xylem. The reason it can manage without them is that:

(A) its small size lets materials enter and leave directly across its surface by diffusion

(B) it does not need oxygen or food to live(C) it is too primitive to require any transport

(D) it keeps blood inside one giant cell

56. Why does a large multicellular plant or animal NEED a transport system, while a tiny organism does not?

(A) its inner cells are too far from the surface to be supplied by simple diffusion alone

(B) it has no surface across which anything can pass

(C) it never needs oxygen, food or water (D) it can only survive while floating in water

57. Which row correctly matches each tube with what flows through it?

(A) ureter → food; xylem → urine; phloem → water

(B) ureter → urine; xylem → water and minerals; phloem → food

(C) ureter → blood; xylem → food; vein → urine

(D) artery → urine; vein → water; phloem → blood

58. Both the human kidney and a plant's transpiration involve the movement of:

(A) haemoglobin (B) water

(C) platelets (D) bone

59. Comparing the two transport systems, which analogy is the most reasonable?

(A) the plant's xylem is like the veins carrying blood back to the heart

(B) the plant's phloem works like the kidneys filtering waste

(C) a plant's root hairs act like the heart's ventricles

(D) the heart pumps blood through vessels in animals, much as the transpiration pull draws water up the xylem in plants — but plants have no pump

60. Which overall summary of transport in living things is fully correct?

(A) in animals the heart pumps blood through vessels; in plants the xylem carries water up and the phloem carries food

(B) in animals the xylem pumps blood; in plants the heart carries water

(C) in animals veins carry blood away from the heart; in plants phloem carries water up

(D) both animals and plants use a heart to move all their materials

Solutions — Science (Class 7), Chapter 11: Transportation in Animals and Plants — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	D	31	A	41	D	51	C
2	C	12	C	22	A	32	D	42	A	52	A
3	A	13	B	23	B	33	B	43	C	53	D
4	B	14	B	24	A	34	A	44	C	54	C
5	B	15	D	25	C	35	D	45	B	55	A
6	A	16	C	26	C	36	B	46	B	56	A
7	A	17	C	27	D	37	D	47	B	57	B
8	D	18	B	28	B	38	A	48	C	58	B
9	D	19	C	29	A	39	C	49	C	59	D
10	A	20	D	30	C	40	B	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Stopping bleeding is clotting, the job of platelets, so a poor clotting time with normal oxygen and immunity points to low platelets. Red blood cells carry oxygen, which is stated to be normal. White blood cells fight germs, also stated to be normal. Plasma is mostly the fluid that carries everything; a clotting failure is specifically a platelet shortfall, not a fluid problem.

2. (C) Tissues being starved of oxygen points straight to the oxygen-carrier: haemoglobin inside red cells. Too little of it means less oxygen delivered, causing tiredness and breathlessness. Too few platelets affects clotting, not oxygen supply. Too few white cells affects defence against germs. Excess plasma is not the textbook cause of anaemia; the oxygen-carrying pigment is.

3. (A) Plasma is the straw-coloured liquid in which cells float and in which dissolved substances — digested food, urea and others — are carried around the body. Making haemoglobin is not a plasma job; haemoglobin sits inside red cells. Destroying bacteria is the white cells' work. Sealing wounds is done by platelets, not the fluid itself.

4. (B) A mature red cell has no nucleus, which frees up internal space that is packed with haemoglobin — the more haemoglobin, the more oxygen carried. Fighting germs is the white cells' role, not red cells'. Clotting is the platelets' job. Red cells do not make food; that is a plant-leaf process, not a blood-cell one.

5. (B) White cells fight infection, so during a bacterial attack the body produces more of them to overwhelm the germs. Carrying oxygen is the red cells' job, unrelated to a germ count. Clotting is done by platelets, not white cells. White cells do not transform into red cells; the cell types are distinct and made separately.

6. (A) Red cells carry oxygen using haemoglobin; white cells defend against disease; platelets help clotting. The other rows scramble these jobs — for example assigning oxygen to white cells or defence to red cells — which contradicts each component's known function.

7. (A) When blood settles, the cells (heavier) sink and the pale-yellow liquid — plasma — rises on top. Haemoglobin is not a separate liquid layer; it is inside the red cells that packed at the bottom. Urea is a dissolved waste within plasma, not a layer by itself. Lymph is a separate tissue fluid, not what you get by spinning a blood sample.

8. (D) Oxygen delivery depends on haemoglobin inside red cells, so restoring red cells most directly restores oxygen transport. Platelets help clotting, not oxygen carriage. White cells handle germs. Plasma carries dissolved materials but is not the oxygen carrier; the red cells are.

9. (D) Haemoglobin is an iron-rich pigment, so dietary iron is needed to build it and keep oxygen transport healthy. Platelets are not made of iron. Plasma is mostly water with dissolved substances, not iron. White cells are not built from iron either; the iron link is specifically to haemoglobin.

10. (A) Oxygen-rich blood is bright red and oxygen-poor blood is a duller, darker red — both because of haemoglobin's colour with and without oxygen. Platelet count does not change the redness like this. Blood does not turn into urine; that is a kidney waste, not a blood colour. Blood is never actually blue, so the 'blue blood' idea is a myth.

11. (D) Arteries carry blood freshly pumped from the heart at high pressure, so each heartbeat sends a surge — the spurting. Carrying blood back toward the heart is the vein's job, and that flow is low-pressure and steady. Valves are found in veins, not the cause of arterial spurting. Arteries are actually thick-walled, not thinner than veins.

12. (C) Valves, thin walls and low pressure are the signature of veins, which return blood to the heart and use valves to stop backflow. Arteries have thick walls, high pressure and no valves. Capillaries are one-cell-thin exchange vessels without valves. A ureter is not a blood vessel at all; it carries urine.

13. (B) Vein blood travels at low pressure on its way back to the heart and could otherwise drift backward, so one-way valves keep it heading the right way. The 'very high pressure' claim is backwards — that describes arteries, which is exactly why arteries do not need valves. Arteries are not pumps, and valve action does not create the heartbeat sound (the heart's own valves do).

14. (B) Capillary walls are just one cell thick, a tiny barrier across which oxygen, food and wastes pass easily between blood and body cells. Thick muscular walls describe arteries and would block exchange. Valves belong to veins and are not an exchange feature. Capillary walls are not cartilage; they are thin living cells.

15. (D) Blood leaves the heart through an artery, slows into thin capillaries where exchange with the muscle happens, then collects into a vein heading back to the heart. Starting with a vein reverses the direction away from the heart. Putting the vein before capillaries skips the exchange step in the right place. Capillaries are the middle exchange stage, not the start.

16. (C) The rule of thumb (arteries oxygen-rich, veins oxygen-poor) holds for most vessels, but the lung vessels are the exception: the artery going to the lungs carries oxygen-poor blood and the vein returning carries oxygen-rich blood. So it is not exception-free. Saying arteries 'always' carry oxygen-poor blood is wrong — most carry oxygen-rich. Age has nothing to do with it.

17. (C) Arteries take blood at high pressure straight from the heart, so thick elastic muscular walls let them stretch with each surge and not burst. Arteries do not hold valves — that is veins. Arterial blood is fast and high-pressure, not slow. Arteries do not filter blood; that filtering is the kidneys' job.

18. (B) Arteries take blood straight from the pumping heart, so pressure is highest there; capillaries are the tiniest, most spread-out vessels, so flow slows right down to allow exchange. Veins are low-pressure, so they are not the highest. Capillaries are not high-pressure. Flow is not slowest in arteries — they carry fast, surging blood.

19. (C) Blood is always red — bright when oxygen-rich, dark when oxygen-poor — and only appears bluish through the skin because of the way light is filtered. It is never truly blue, even before reaching air. Oxygen-poor blood is dark red, not blue. Plasma is pale yellow, not blue.

20. (D) Thick muscular elastic walls are the artery's hallmark, suited to the high pressure of heart-pumped blood; veins have thinner walls. Valves are the vein's feature, not the artery's. A one-cell-thin wall describes capillaries, not arteries. Carrying blood toward the heart is the vein's role, opposite to the artery's.

21. (D) Four chambers keep oxygen-rich and oxygen-poor blood apart, so the blood sent to the body is fully oxygenated and efficient. The arrangement is about separation, not loudness of the beat. The heart is muscular and still works hard regardless. Separation does not 'speed up colour change'; colour depends on oxygen, not chamber count.

22. (A) Ventricles do the powerful pumping that pushes blood out of the heart, so they need thicker muscle. The atria only receive incoming blood gently, so they are thinner-walled. Storing urine is a urinary-bladder function, not a heart chamber's. The pulse is the artery expanding, not a valve inside a ventricle.

23. (B) The lub-dub is the sound of the heart's valves snapping shut as the chambers pump blood. Capillary flow is far too gentle to make that sound. Breathing makes its own sounds, separate from the heartbeat. The wrist pulse is felt, not the source of the chest's lub-dub.

24. (A) The pulse is the rhythmic expansion of an artery as each heartbeat pushes blood through it, felt at the wrist. Clotting seals wounds and has nothing to do with the wrist throb. Vein valves do not cause an arterial pulse. The lungs handle air, not a throb in the arm artery.

25. (C) If 90 beats occur in 60 seconds, then in 15 seconds (one quarter) there are $90 \div 4 \approx 22$ to 23 beats, with one pulse per beat. 90 in 15 seconds would be the full minute's count crammed into a quarter — far too high. 45 would be half a minute's worth. 6 is far too few for that rate.

26. (C) 20 seconds is one-third of a minute, so multiply by 3: $24 \times 3 = 72$ beats per minute, which is also a healthy resting rate. 24 is just the count in 20 seconds, not per minute. 48 would be multiplying by 2, the wrong factor. 120 over-multiplies; it does not match 24 in 20 seconds.

27. (D) Hard exercise makes muscles demand more oxygen and food and produce more waste, so the heart beats faster to deliver and remove these quickly. 'Less oxygen' is the opposite of what active muscles need. Clotting is unrelated to exercise heart rate. A faster beat is not 'slower' pumping, so the cooling-by-slowing idea is self-contradictory.

28. (B) A typical resting pulse is around 72, so 70 is normal but 130 is far above that — much higher than usual at rest. It is not 'exactly normal,' since it nearly doubles the typical value. It is certainly not lower. Such a rate is possible (for example when anxious or unwell), so it is not impossible.

29. (A) The heart is a rhythmically contracting muscular pump that drives blood through the vessels. It is not bone; it is muscle. Filtering wastes is the kidneys' job, not the heart's. Making food and using phloem are plant processes, completely outside the heart's role.

30. (C) Urea is the chief nitrogenous waste that kidneys filter from blood into urine. Glucose is a useful food the body keeps, not a waste to dump. Oxygen is a gas carried by red cells, not a urine waste. Haemoglobin stays inside red cells; losing it in urine would mean disease, not normal excretion.

31. (A) Nephrons are the tiny filtering units packed inside each kidney. Stomata are leaf pores in plants, not in kidneys. Platelets are blood cells for clotting, not filters. Ventricles are heart chambers, not kidney units.

32. (D) Urine is made in the kidney, runs down a ureter to be stored in the bladder, then leaves through the urethra. Putting the urethra before the bladder reverses the exit. Starting at the bladder ignores that urine is formed in the kidney first. Starting at the ureter skips the kidney where urine is actually made.

33. (B) Sweat carries out water and dissolved salts, so it is a form of excretion as well as a cooling mechanism. The skin does not breathe in oxygen as its main job. Carrying food is the blood's role, not sweat's. Sweating does not thicken blood for clotting; that is the platelets' function.

34. (A) Dialysis uses a machine to filter wastes from the blood when the kidneys cannot. Transpiration is water loss from plant leaves, unrelated. Transfusion is the transfer of donated blood, not waste cleaning. Circulation is the normal movement of blood, not a treatment for kidney failure.

35. (D) Urea is poisonous if it accumulates, so the kidneys must keep clearing it from the blood. It is a waste, not a food worth storing. Blood colour comes from haemoglobin, not urea. Clotting depends on platelets, not on urea.

36. (B) Urea forms in the body's cells, travels dissolved in the blood, is filtered out by the kidneys and finally leaves dissolved in urine. Xylem and stomata are plant tissues, not a path for human urea. Urea is not made in the bladder nor exhaled by the lungs. Urea is not 'made in urine' and is never stored in the heart.

37. (D) Excretion is the removal of waste and excess water, which both sweating and urine formation accomplish. Bringing in oxygen is the opposite of excretion (it is intake). Delivering food is transport by blood, not waste removal. Pumping blood is circulation, a different process from excretion.

38. (A) Some plant wastes collect in leaves that eventually drop off, carrying the waste away — a genuine plant strategy. Ureters and bladders are animal urinary parts; plants have none. Plants have no blood or nephrons to filter. Root hairs absorb water, they do not sweat out salts.

39. (C) Gums and resins are among the waste materials plants store and may release, for example from the trunk. Food for the roots travels inside the phloem, not as oozing resin.

Water rising to leaves moves inside the xylem, not as resin. Oxygen is a gas exchanged through stomata, not a sticky resin.

40. (B) Stomata are the leaf pores through which gases and water vapour pass in and out, including some waste release. Root hairs are underground and absorb water; they do not release leaf gases. Nephrons are kidney units in animals. Platelets are blood cells, not plant pores.

41. (D) Plants produce less harmful waste than animals and can store some in tissues or shed it slowly, so their excretion is slower and simpler — no dedicated organs needed. Plants have no kidneys, let alone many. They have no heart or blood vessels. They do make waste, so saying they make none is false.

42. (A) Plants use several routes — stomata for gases and vapour, and storage in leaves, bark, gums and resins — to handle waste. They have no kidneys. Xylem carries water up, not waste out the roots. Plants certainly do produce waste, so the 'no waste' claim is false.

43. (C) Xylem is the tissue that conducts water and minerals upward from roots to leaves. Phloem carries food, not the upward water stream. Stomata are pores for gas/vapour exchange, not transport tubes. Root hairs absorb water from the soil but do not carry it all the way up.

44. (C) Root hairs are fine outgrowths that vastly increase the surface area touching soil water, boosting absorption. They are underground and not green, so they do not make food. Plants have no muscles to pump water. Carrying food down is the phloem's job, not the root hairs'.

45. (B) Evaporation of water from leaves (transpiration) creates a suction — the transpiration pull — that draws water up the xylem. Plants have no heart, so 'heartbeat' is meaningless here. Clotting is a blood process, not a water-moving force. Food movement is in the phloem, not the source of the upward water pull.

46. (B) Heat, dryness and wind all speed up evaporation from leaves, so transpiration increases and water is pulled up faster. Heat does not seal leaves shut and stop it. Transpiration does not reverse to push water downward. Faster transpiration plainly does affect water movement, so 'no effect' is wrong.

47. (B) On a hot afternoon transpiration can outpace the roots' supply even in moist soil, so the plant temporarily loses more water than it takes up and wilts. Moist soil is not poisoned. Xylem does not transform into phloem. A short wilt does not mean food-making has permanently stopped; the plant usually recovers in the cool.

48. (C) Root hairs are the main absorbers of soil water and minerals, so losing them most directly hurts absorption. Food-making in leaves depends on light and chlorophyll, not directly on root hairs. Water loss through stomata happens in the leaves, not via root hairs. Phloem food transport is a separate tissue and process.

49. (C) Xylem carries water upward, so a blockage starves the leaves of water first — they would wilt. Food, not water, is the phloem's cargo, so 'food can't reach the leaves' misnames the tissue. The roots' problem would be food, which travels in the phloem, not the blocked xylem. The 'roots losing minerals' idea reverses the upward direction of xylem flow.

50. (D) Phloem transports food from the leaves to the rest of the plant, including downward to roots. Xylem carries water and minerals upward, not food. Stomata are gas pores, not transport tubes. Root hairs absorb water; they do not distribute food.

51. (C) Phloem can move food both upward and downward, delivering it to growing tips, fruits and roots as needed. Restricting it to only-up or only-down misses that food goes wherever it is needed. Food certainly does move in plants — that is the whole point of the phloem.

52. (A) Removing the phloem ring cuts the downward food path, so the part below the ring (toward the roots) is starved of food. Water still rises in the intact xylem, so the upper part keeps its water. The leaves can still photosynthesise; only the delivery downward is blocked. The xylem is inside the wood, not in the bark ring, so it is not the cut tissue.

53. (D) Phloem can carry food in several directions at once, so a leaf can supply both a downward root tip and an upward fruit together. It is not limited to upward-only flow. Food is not handed off into the xylem, which carries water. Phloem transport is not switched on only in sunlight; food moves whenever needed.

54. (C) Xylem moves water and minerals up; phloem distributes food in multiple directions — the correct division of labour. Swapping their cargoes (food in xylem, water in phloem) is the classic mix-up and is wrong. They do not both carry only water. Oxygen and urea are animal-transport items, not the named plant cargoes.

55. (A) An Amoeba is tiny, so oxygen, food and wastes can move directly across its surface by diffusion — no transport system needed. It still needs oxygen and food like any living thing. 'Too primitive' is not a scientific reason; size and diffusion are. A single cell holds no blood to circulate.

56. (A) In a big body, the inner cells are far from the surface, so diffusion alone is too slow — a transport system is needed to reach them. Large bodies do have surfaces, but diffusion cannot reach deep enough. Large organisms certainly need oxygen, food and water. Living on land or in water does not decide whether a transport system is needed; size does.

57. (B) A ureter carries urine, xylem carries water and minerals, and phloem carries food — each tube with its true cargo. The other rows scramble these, for example sending food through a ureter or urine through the xylem, which contradicts what each tube actually transports.

58. (B) Kidneys remove excess water (in urine) and transpiration is the loss of water vapour from leaves, so water is the shared theme. Haemoglobin is only in animal blood, not in plant transpiration. Platelets are blood cells, not part of either water process. Bone has nothing to do with kidney filtration or leaf water loss.

59. (D) Both systems move materials around the body, but plants rely on the transpiration pull rather than a pump like the animal heart — a fair, careful analogy. Xylem carries water UP and is not like blood-returning veins. Phloem carries food, not filtered waste like kidneys. Root hairs absorb water and are nothing like pumping ventricles.

60. (A) Animals use a pumping heart and blood vessels; plants use xylem (water up) and phloem (food) — the correct combined picture. Animals have no xylem and plants have no heart, so swapping them is wrong. Veins carry blood TOWARD the heart, and phloem carries food, not water — so that option mislabels both. Plants have no heart, so 'both use a heart' is false.